Advanced Topics in Computer Graphics I

Universität Bonn
Institut für Informatik II
May 7, 2026

Summer term 2026
Prof. Dr. Reinhard Klein

Domenic Zingsheim

Sheet R04 - Whitted-Raytracing

Hand in your solutions via eCampus by Fr, 15.05.2026, **08:00 a.m.**. Compile your solution to the theoretical part into a single printable PDF file. For the practical part, hand in a single ZIP file containing only the exercise* folder within the src/ directory. Please refrain from sending the entire framework.

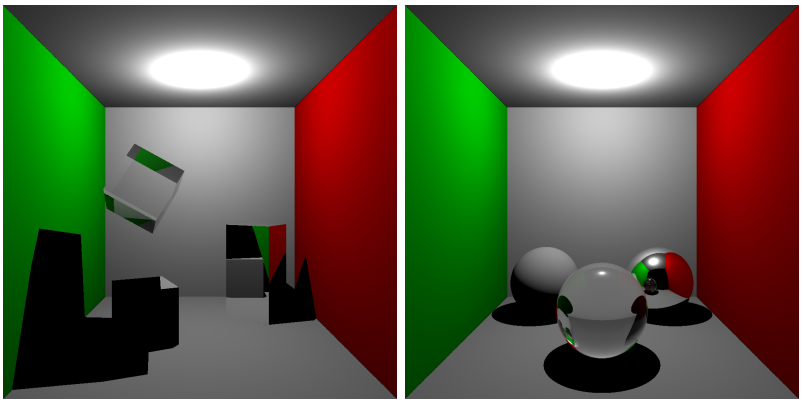

Figure 1: Cornell Box scene rendered with Whitted raytracing. To obtain these renderings, run `./bin/exercise04_WhittedRaytracing -s data/exercise04_WhittedRaytracing/cornell_box_cubes.xml` and `./bin/exercise04_WhittedRaytracing -s data/exercise04_WhittedRaytracing/cornell_box_spheres.xml` from the project root, respectively.

Assignment 1) Whitted-Raytracer *(8 Pts)*

In this exercise, we implement a raytracer based on the Whitted algorithm[1]. The framework is similarly structured to the last exercise. We now use the `XMLSceneLoader` to load scene representations from xml files instead of defining them manually in the C++ code. The ray generator that implements the Whitted raytracing algorithm is defined in the `whittedraygenerator.{cpp,cu,cuh,h}` files. A point and directional light source is implemented in the `lightsources.{cpp,cu,cuh,h}` files. An opaque BSDF modeling diffuse and perfect specular reflections, as well as a refractive BSDF, modeling reflection and transmission at the interface between two media (e.g. air and glass) is implemented in the `bsdfmodels.{cpp,cu,cuh,h}` files.

a) Compute the diffuse reflections of the light sources on opaque materials in `whittedraygenerator.cu`. Generate a ray towards to light source using the `EmitterVPtrTable::sampleLight()` method. Determine the visibility of the light source using the `traceOcclusion()` method. Evaluate the BSDF at the surface interaction towards the light direction using the `BSDFVPtrTable::evalBSDF()` method. Accumulate the contribution of the light source in the `output_radiance` variable. Note that only diffuse BSDF components contribute here, since the probability that a given light direction corresponds to a perfect specular reflection is zero.

4 Pts

b) In `bsdfmodels.cu`, generate a reflection ray for the specular component of the opaque BSDF in `__direct_callable__opaque_sampleBSDF()`, as well as reflection and transmission rays for the refractive BSDF in `__direct_callable__refractive_evalBSDF()`.
Hints: The surface normals point outwards. You can use Schlick's approximation for the Fresnel term to compute the amount of light reflected or transmitted

4 Pts

If you finished all tasks successfully your results should look like the ones in Figure 1.

Theoretical Assignment

Assignment 2) Refraction in Wedge *(3Pts)*

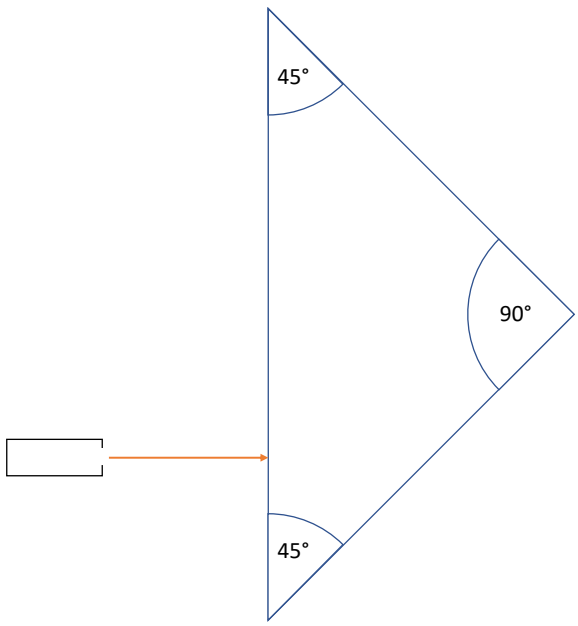

Figure 2: Wedge illuminated by a laser.

Consider the scene depicted in Figure 2 in which a laser is shooting unpolarized light rays into a wedge which is surrounded by air (refractive index $n_a = 1$). The light rays are hitting the wedge perpendicular to the surface.

a) Assuming the wedge has a refractive index of $n_w = \sqrt{3/2} \approx 1.225$, draw all light paths for the laser light into the scene (with correct angles!).
b) Consider the same scene but with a wedge consisting of a different material. The new material has a refractive index of $n'_w = 5$. Draw the new light paths into the scene.
c) Describe the difference between the two scenarios.
d) Calculate $\hat{n}_w$ which is the smallest refractive index, that leads to the same result as n'_w.

Assignment 3) Refraction in Slab *(2Pts)*

Consider the scene depicted in Figure 3 in which a laser is shooting unpolarized light rays onto a glass slab (refractive index $n_g = 1.5$) surrounded by air (refractive index $n_a = 1$) at an angle of 50 degrees

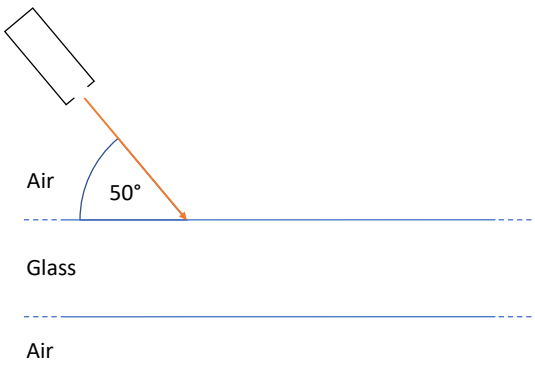

Figure 3: Slab illuminated by a laser.

measured to the surface. Assume that the glass slab has parallel interfaces at the top and bottom, and extends infinitely to both sides in Figure 3.

a) Draw the light paths (using correct angles) of the laser light in the scene. Since there is an infinite number of paths, it is enough to only draw paths up to 6 reflections/refractions.

b) Calculate how much light is emitted to the top and how much light is emitted to the bottom of the slab as a fraction of total light emitted by the laser.

Hints:

- Use the properties of a geometric series.
- The light polarization does not change between the top and bottom interface of the slab.

References

[1] Turner Whitted. An improved illumination model for shaded display. In *Proceedings of the 6th annual conference on Computer graphics and interactive techniques*, page 14, 1979.

Good luck!